

SKILLS

Languages: Rust, Python, TypeScript, Java, C++, Gleam
ML/AI: Keras, PyTorch, Hyper-parameter optimization, RAG, LLM evaluation, Agentic systems, Google ADK
Frameworks: React, Flask, Litestar, FastAPI
Databases: LanceDB, MySQL, MongoDB, DynamoDB
Cloud: Google Cloud (Compute Engine, Cloud Storage, Vertex AI), AWS (EC2, S3, DynamoDB)

EMPLOYMENT

LexisNexis Legal & Professional Raleigh, NC
Senior Data Scientist I May 2024 - Present

- Led the migration from RAG to an agentic legal drafting system as tech lead, and own its continued development.
- Reduced customer-facing product runtime by 24.8% and cut time spent at peak memory from 86.4% to 68% via lazy module loading, LLM call profiling, and parallelization.
- As part of a 3-person team, improved complaint drafting usefulness (as rated by attorney reviewers) from 40% to 86% and motion drafting from 59.2% to 76%.
- Developed a fast (median 12ms) passage filtering approach based on contrastive loss.
- Mentored an intern on a large-scale, full factorial experiment investigating Pareto-optimal configurations for draft quality and latency.
- Built team developer tooling: a Rust MITM proxy with request replay, a web UI for visualizing intermediate agent steps, and custom Python lint rules beyond `ruff`'s rule set.
- **Technology:** Python, Litestar, Google ADK, React, Rust, TypeScript, Tailwind

Amazon New York, NY / Bellevue, WA
Software Dev Engineer Intern May 2023 - Aug 2023

- Implemented and shipped profile locks for Prime Video on Echo Show devices.
- **Technology:** React Native, TypeScript

Software Dev Engineer Intern May 2022 - Jul 2022

- Developed a full-stack system to publish announcements in scorecards used by delivery service partners.
- **Technology:** React/Redux, TypeScript, Redux Saga, DynamoDB, Java Spring

HONORS

Distinguished Reviewer, ICSE 2026 Mar 2026
Google Developer Expert - ML and GCP Mar 2025 - Present
A global network of experts recognized by Google for deep technical expertise, community leadership, and mentorship.

Google Cloud Champion Innovator - Cloud AI/ML Oct 2022 - Mar 2025
A Googler-nominated global network of 500+ technical experts in Google Cloud; merged into the Google Developer Experts program in 2025.

Google Cloud Research Innovator Feb 2022
Research Innovators are a global community of researchers driving scientific breakthroughs with Google Cloud.

RELEVANT PROJECTS

pi-permission-system (maintained fork) May 2026 - Present
TypeScript, Node.js [GitHub](#)
Maintain a permission-system fork for the Pi coding agent. Redesigned shell permissions using grammar-based analysis of pipelines, substitutions, and redirections, with protected-path checks and argument-aware approval rules. Extended enforcement to background commands and fixed concurrent subagent approval handling.

Q&A System for Zotero Oct 2024 - Present
Rust, SQLite, LanceDB [GitHub](#) :: [crates.io](#)
Built an academic question-answering and semantic search system over a user's Zotero library. Wrote a custom PDF parser for academic papers (p50 $\approx$ 2,400 pages/s, p90 $\approx$ 900 pages/s; M4 Pro) and a RAG library with support for multiple providers and tool calling.

JournalBear

Jan 2017 - Jul 2026

TypeScript, React, Zustand, Electron, SwiftUI

[GitHub](#) :: [App Store](#) :: [Softpedia](#)

Cross-platform journaling app with AES-256 encryption and separate native apps for iOS and macOS. Rated 4/5 by Softpedia.

The 1 Billion Row Challenge (1BRC)

Feb 2026 - Apr 2026

Zig

[GitHub](#)

Wrote a Zig implementation for the 1BRC, running in 1.20s (M4 Pro, 14 core) using mmap, open addressing (Wyhash + Robin Hood probing), and packed SIMD.

Programmable Resumes

Aug 2023 - Mar 2025

Python

[GitHub](#)

Developed a specification and implementation for modular, customizable resumes with support for two popular LaTeX templates. Wrote this resume using this tool.

EDUCATION

North Carolina State University

Raleigh, NC

Ph.D. Computer Science

Aug 2019 - Jul 2024

- Advisor: *Dr. Tim Menzies*
- Dissertation: [Guidelines for the Application of Neural Technologies in Software Analytics \(or: How to Do More with Less in SE\)](#)

PES University

Bangalore, India

B.E. Computer Science – Advisor: Dr. Snehanshu Saha

Aug 2015 - May 2019